

BioSync

Group 2

CSC 289 — Programming Capstone, Spring 2026

Repository URL: <https://github.com/CSC-289-Biometric-Group-2/BioSync-Group2>

Date of final submission

BioSync is a biometric tracking application that brings data collection and analysis to one place.

This application is intended for those in need of keeping track of medical information and vitals. This could include health-conscious individuals or those with chronic illnesses that need to have records of this data and what it could mean. Caregivers such as doctors or nurses can also utilize BioSync to access patient health data.

Currently, health tracking is scattered across different devices and apps. BioSync doesn't specialize in tracking a specific dataset, but a multitude of metrics such as heartrate, blood pressure, blood oxygen level, and more.

BioSync is currently in a semi-usable state, and most of the application features are present in its latest build, not including external tracking devices.

Team Roster & Roles

For each team member, include:

- Name
- Primary role(s) during the project (e.g., frontend lead, Scrum Master rotations, DB specialist)
- GitHub username
- One sentence describing their main contribution

If your team had a rotating Scrum Master, note who served in which sprint.

My Notes:

Sarah Ayres

- Primary role(s): Product Owner, Scrum Master
- GitHub: Sayres-dev
- Main Contribution(s):

Joshua Eckman

- Primary role(s): Developer
- GitHub: Eckmanj8966
- Main Contribution(s): Application initialization, and more thorough heartrate thresholds.

Myra Williams

- Primary role(s): Scrum Master, Developer
- GitHub: mwill2214
- Main Contribution(s):

Toni Wilson

- Primary role(s):
- GitHub: toniwilson
- Main Contribution(s):

Product Vision

Estimated length: 2–3 pages

Where it comes from: Product brief, pitch presentation, persona work.

This is the expanded version of your project overview. Include:

4a. Target Users / Personas — Describe 2–3 representative users. For each, give a name, a brief backstory, and what they need from your application. These can be the personas from your pitch, cleaned up.

My Notes:

Representative User 1:

Joy Edwards is a busy grad student who is interested in staying on top of her health as well as her grades. She uses many different fitness tracking devices and apps to record her vitals and reach her health goals. BioSync would store all this data in one place and take the stress out of prioritizing her wellness.

Representative User 2:

Jason Spencer is the caregiver of an elderly patient and would benefit from having his patient's vitals recorded between visits. His elderly patient is not technologically inclined and does not know how to record this information into apps by herself. With BioSync, medical data will be gathered and recorded when Jason isn't with his patient and is readily available for him to access whenever he needs it.

Representative User 3:

Micah Davis has a chronic illness that requires multiple trips to various specialists. Tracking his vitals was difficult enough before he got sick but now that this information is important to provide to his doctors, he is having issues finding an app that keeps track of all the data he needs. BioSync not only tracks and records a variety of health data but is easy to share with his doctors in and out of appointments.

4b. Core Value Proposition — In plain language, what makes your application worth using? What's the one thing it does better than the alternatives (including "doing nothing")?

My Notes:

[Needs to be rewritten]

Our platform is a multi-device, multi-domain biometric ecosystem that flows into one dashboard, collecting and displaying cohesive information, and even giving recommendations based on data collected. [Last section not implemented yet.]

4c. Feature Scope — A high-level list of the features your application supports (or was designed to support). Mark each as:

- **Implemented** — working in the current build
- **Partial** — started but not complete
- **Planned** — designed but not built

Be honest. A clear-eyed scope assessment is more valuable than a fiction of completeness.

My Notes:

1. Various biometric categories such as heartrate, blood pressure, blood oxygen level, etc. [Implemented]
2. Account linking between a user and a caregiver. [Implemented]
3. Importing and exporting health data. [Partial]
4. Recommendations based on collected data [Partial]
5. Biometric tracking device connectivity. [Planned]

If it were to be cut down to three items, they would be the various biometric tracking categories, account linking between a user and a caregiver, and the importing and exporting of health data.

Guiding questions:

- Would your personas actually use this? What would make them come back?
- If you had to cut the feature list to three items, which three survive?

User Stories

Estimated length: 2–3 pages

Where it comes from: Sprint planning sessions, product backlog.

List your user stories in priority order. Use standard format:

As a [type of user], **I want to** [action] **so that** [benefit].

Group stories by feature area or sprint. For each story, indicate its status (done, in progress, not started) and which sprint it was worked on.

Important: User stories use “As a user” or “As a [specific persona name]” — never “As a developer” or “As a team.” Technical tasks are not user stories; they belong in your sprint summaries or architecture section.

Guiding questions:

- Does every story describe something a *user* experiences, not something you *built*?
- Are your stories small enough that each one was completable in a single sprint?

My Notes:

As a user, I want to create an account with a unique username so that I can begin using the app. [Done] [Sprint 1]

As a user, I want to be able to securely log in to the application with a created account so that I can catalog my health biometrics. [Done] [Sprint 1]

As a user, I want to be able to select, on account creation, whether I am an individual user or a caregiver so that I can gain access to the features I need. [Done] [Sprint 2]

As a user, I would like to be able to link a caregiver account to my patient profile so that they can view and manage my biometric information. [Done] [Sprint 2]

As a user, I want to be able to update my profile information as necessary so that it properly reflects accurate information. [Done] [Sprint 2]

As a user, I want to receive alerts for concerning readings so that I can seek medical assistance if need be. [Done] [Sprint 2]

As a user, I want to be able to upload official records to my profile to help with my assessment and to store and compare long term patterns. [Done] [Sprint 2]

System Architecture

Estimated length: 2–3 pages

Where it comes from: Technical brief, [CLAUDE.md Links to an external site.](#), architecture decisions made during sprints.

Describe how your application is structured. Include:

6a. Tech Stack — List the languages, frameworks, and major libraries your application uses, with a one-sentence note on why each was chosen (or assigned).

6b. Application Structure — A high-level diagram or description of how the pieces fit together. For a Flask application, this might include: how routes map to templates, where business logic lives, how the database is accessed. A simple box-and-arrow diagram (even ASCII art) is fine. A file tree of your project’s key directories is also useful here.

6c. Route Map — A table listing your application’s URL routes, the HTTP method, what the route does, and whether it requires authentication.

Route	Method	Description	Auth Required?
/	GET	Landing page	No
/login	GET, POST	User authentication	No
/dashboard	GET	Main user dashboard	Yes

Guiding questions:

- Could a new developer read this section and know where to find things in the codebase?
- If someone asked “where does X happen in the code?” could you point them to a file?

Data Model

Estimated length: 2–3 pages

Where it comes from: Technical brief, database design work.

7a. Entity Relationship Diagram — Include a diagram showing your database tables and their relationships. This can be generated from a tool ([dbdiagram.io Links to an external site.](#), [draw.io Links to an external site.](#)), exported from your ORM, or drawn by hand and photographed. The key requirement is that it shows: tables, columns (at least primary keys and foreign keys), and the relationships between tables (one-to-many, many-to-many).

(Placeholder: Insert your ERD image or diagram here.)

7b. Data Dictionary — For each table, provide:

Column	Type	Constraints	Description
id	Integer	PK, auto-increment	Unique identifier
username	String(80)	Unique, not null	User's login name

You don't need to document every column of every table exhaustively if your schema is large. Focus on the tables that are central to your application's core features, and note any non-obvious design decisions (e.g., "We store game state as JSON in a text column because...").

7c. Key Relationships — In a sentence or two per relationship, explain how the tables connect and why. Example: "Each user has many sessions. Each session has many responses. This lets us track performance over time at both the session and individual-question level."

Guiding questions:

- If someone needed to write a new query against your database, would this section give them enough context?
- Are there any "surprising" columns or tables that need explanation?

Key Features & Implementation Notes

Estimated length: 3–5 pages

Where it comes from: Sprint work, PR descriptions, code comments.

Pick your 3–5 most important or interesting features. For each one, write a short section (half a page to a full page) covering:

- **What it does** — from the user’s perspective.
- **How it works** — a brief technical explanation. Not a code walkthrough, but enough that a developer could understand the approach. Include a code snippet only if it genuinely clarifies the explanation (e.g., a key algorithm, a non-obvious query).
- **Challenges encountered** — what was harder than expected? What did you try that didn’t work before arriving at the current approach?

This section is where your binder becomes more than a spec document. The “challenges encountered” subsection is where you demonstrate learning.

Guiding questions:

- If you were explaining this feature in a job interview, what would you say?
- What would you do differently if you built this feature again?

Deployment

Estimated length: 1–2 pages

Where it comes from: Sprint 3+ deployment work.

Describe how your application is (or would be) deployed:

- **Hosting platform** (Render, Railway, PythonAnywhere, etc.)
- **Database in production** (SQLite, PostgreSQL, etc.) and any differences from development
- **Environment variables** needed (list the keys, not the values)
- **Deployment steps** — what someone would need to do to deploy a fresh copy

If your application is not yet deployed, describe where you are in the process and what remains.

Guiding questions:

- Could a teammate who wasn't involved in deployment get the app running from this documentation?
- Are there any “gotchas” someone would hit during deployment that aren't obvious?

Lessons Learned & Retrospective

Estimated length: 1–2 pages

Where it comes from: Sprint retrospectives, personal reflection.

This is the most important section of the binder from a learning perspective. Each team member should contribute at least one paragraph reflecting on:

- **What they learned** — a specific skill, tool, or concept they didn't know before.
- **What they'd do differently** — with the benefit of hindsight.
- **What they're proud of** — a specific contribution or moment.

This section can be written as individual reflections (labeled by name) or as a team synthesis — your choice.

Guiding questions:

- What would you tell next semester's students about this course?
- What was harder than you expected? What was easier?

Appendices

Estimated length: As needed (does not count toward the 20–30 page target).

These are optional. Include only what adds value.

Possible appendices:

- **A. Wireframes or mockups** — early design artifacts, especially if they show how the design evolved.
- **B. Pitch deck or presentation slides** — a link or embedded copy of your pitch materials.
- **C. API documentation** — if your application exposes an API, document the endpoints here.
- **D. Meeting notes or standup logs** — selected examples showing team communication.
- **E. [CLAUDE.md Links to an external site.](#)** — your project’s AI assistant context file, if you used one.
- **F. Design assets received** — brand guide, color palette, or other materials from design partners.